INCH-POUND
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PERFORMANCE SPECIFICATION

HOSE, AIR DUCT: HIGH-TEMPERATURE FLEXIBLE, REINFORCED

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers reinforced flexible hose used for transferring heat to crew compartments and preheating engine compartments, oil pans, batteries, fuel and coolant lines, and other components to assist in starting engines under extreme cold weather conditions (see 6.1).

1.2 Classification. The hose is classified as follows (see 6.2 and 6.3).

1.2.1 Types. The types of hoses are as follows:

Type I	Low working pressure
Type II	Medium working pressure

1.2.2 Classes. The classes of hoses are as follows:

Class 1	Operating temperature range: -65 to +300 degrees Fahrenheit (°F).
Class 2	Operating temperature range: -65 to +450 °F.
Class 3	Operating temperature range: -65 to +600 °F.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: U.S. Army Tank-automotive and Armaments Command, ATTN: AMSTA-TR-E/AEIT, Warren, MI 48397-5000, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 4720

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2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirement documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Non-Government publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DoDISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS are the issue of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D380	- Rubber Hose (DoD Adopted).
ASTM D1171	- Standard Test Method for Rubber Deterioration - Surface Ozone Cracking Outdoors or Chamber (Triangular Specimens) (DoD Adopted).
ASTM G21	- Standard Practice for Determining Resistance of Synthetic Polymeric Materials to Fungi (DoD Adopted).

(Application for copies should be addressed to the American Society for Testing and Materials, 100 Barr Harbor Drive, West Conshohocken, PA 19428-2959.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2 and 6.4), a sample shall be subjected to first article inspection in accordance with (IAW) 4.1.1.

3.2 Design, materials, and manufacturing processes. The design, materials, and manufacturing process selection is the prerogative of the contractor as long as all articles submitted to the Government fully meet the operating, interface, support and ownership, and operating environment requirements specified.

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3.2.1 Recycled, recovered, or environmentally preferable materials. Recycled, recovered, or environmentally preferable materials should be used to the maximum extent possible provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

3.3 Operating requirements. Each hose shall provide the following functional, operational, and performance requirements as specified in 3.3.1 through 3.3.5.

3.3.1 Leakage. The hose shall not leak air at a rate of more than 0.02 cubic feet per minute (ft^3/min), per inch (in.) of inside diameter (ID), per foot (ft) of length, at the applicable maximum positive working pressure specified in table I.

TABLE I. Working pressure requirements.

Hose ID, in.		Working pressure, pounds per square inch gage (psig)			
		Type I		Type II	
Minimum (min.)	Maximum (max.)	Negative	Positive	Negative	Positive
0.50	3.00	-5.0	5.0	-5.0	16.0
3.01	4.00	-2.0	4.0	-2.0	12.0
4.01	5.00	-1.0	3.0	-1.0	10.0
5.01	6.00	-0.5	2.5	-0.5	8.0

3.3.2 Proof and burst pressures. The hose shall withstand the applicable proof and burst pressures specified in table II without rupturing.

TABLE II. Proof and burst pressure requirements.

Hose ID, in.		Proof pressure, psig		Burst pressure, psig	
Min.	Max.	Type I	Type II	Type I	Type II
0.50	3.00	10.0	32.0	20.0	64.0
3.01	4.00	8.0	24.0	16.0	48.0
4.01	5.00	6.0	20.0	12.0	40.0
5.01	6.00	5.0	16.0	10.0	32.0

3.3.3 Vacuum resistance. While curved to a radius equal to five times the outside diameter (OD) of the hose, the hose shall withstand a vacuum of 10 in. of mercury without the OD being reduced more than 5 percent (%), or without the ID collapsing or separating from the balance of the hose structure.

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3.3.4 Flexibility. The hose shall withstand a bending of 180 degrees (°) on four points spaced 90° apart, around a rigid cylinder with a diameter equal to the ID of the hose, without permanent kinking, flattening, fracture, or deformation in excess of 2%.

3.3.5 Collapsing resistance. The hose shall withstand a minimum load of 50 pounds (lb), gradually administered within a time period of 15 seconds maximum. The 50-lb load shall be applied perpendicular to the axis of the hose for a minimum of 5 minutes, without permanent setting, distortion, or deformation in excess of 5%.

3.4 Interface requirements. Each hose shall accommodate the interface requirements as specified in 3.4.1 and 3.4.2.

3.4.1 Envelope and interface dimensions. The overall envelope and interface dimensions shall be as specified on the interface drawing(s) (see 6.2).

3.4.2 Interchangeability. All hoses shall be interchangeable with parts of other manufacturers having the same military part number.

3.5 Support and ownership requirement. Unless otherwise specified (see 6.2), identification marking shall be permanent, legible, and shall include as minimum the following:

a. Nomenclature “HOSE, AIR DUCT”.
b. Military part number.
c. Manufacturer’s brand or firm name, or CAGE number.
d. Manufacturer’s serial number.
e. Date of manufacture (month and year).
f. Contract or order number.
g. Symbol “US”.

3.6 Operating environment requirements. Each hose shall operate under the following environmental conditions as specified in 3.6.1 through 3.6.5, and with the exception of 3.6.5, shall subsequently meet the performance requirements of 3.3.1 and 3.3.2.

3.6.1 High temperature resistance. The hose shall not crack, break or become brittle when subjected to the applicable maximum temperature for its class (see 1.2.2).

3.6.2 Low temperature resistance. The hose shall not crack, break or delaminate when subjected to temperatures down to -65°F.

3.6.3 Vibration resistance. The hose shall not crack, break or delaminate when subjected to a constant 0.060-in. double-amplitude vibration, at a frequency cycled between 10 and 55 hertz (Hz) in one-minute cycles, applied both parallel and perpendicular to the hose axis for a minimum of 50 hours in each direction.

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3.6.4 Fungus resistance. The hose shall show no evidence of fungal growth when subjected to environments conducive to spore germination and growth (air temperature between 35 and 100°F and relative humidity between 60 and 100%).

3.6.5 Ozone resistance. The hose shall not crack or break when subjected to a minimum of 72 hours of ozone exposure at 104 ± 2°F, with a partial ozone pressure of 50 ± 5 millipascals (mPa).

4. VERIFICATION

4.1 Classification of inspection. The inspection conditions specified herein are classified as follows:

a. First article inspection (see 4.1.1).
b. Conformance inspection (see 4.1.2).

4.1.1 First article inspection. First article inspection shall be performed on the first nine production-representative units of an order when a first article sample is required (see 3.1). Inspection shall include all the verifications listed in table III and shall be in the order as specified in table IV.

TABLE III Verification methods.

Title	Requirement	Verification
Operating requirements	3.3	4.3.3
Leakage	3.3.1	4.3.3.1
Proof & burst pressures	3.3.2	4.3.3.2
Vacuum resistance	3.3.3	4.3.3.3
Flexibility	3.3.4	4.3.3.4
Collapsing resistance	3.3.5	4.3.3.5
Interface requirements	3.4	4.3.4
Envelope and interface dimensions	3.4.1	4.3.4.1
Interchangeability	3.4.2	4.3.4.2
Support and ownership requirement	3.5	4.3.5
Operating environment requirements	3.6	4.3.6
High temperature resistance	3.6.1	4.3.6.1
Low temperature resistance	3.6.2	4.3.6.2
Vibration resistance	3.6.3	4.3.6.3
Fungus resistance	3.6.4	4.3.6.4
Ozone resistance	3.6.5	4.3.6.5

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4.1.2 Conformance inspection. Conformance inspection shall include the verifications as specified in table III as defined in the contractual sampling plan (see 6.2 and 6.5).

4.2 Order of inspection. First article inspection sequence shall be as specified in table IV. Other testing may be in any order except that the environmental requirements verifications (see 4.3.6) shall precede the leakage test (see 4.3.3.1) and the proof and burst pressure tests (see 4.3.3.2).

TABLE IV. Order of first article inspection.

Sample number	Title
1	Flexibility Collapsing resistance
2	Low temperature resistance Leakage Proof & burst pressures
3	High temperature resistance Leakage Proof & burst pressures
4	Envelope and interface dimensions Interchangeability Support and ownership Vacuum resistance
5	Vibration resistance Leakage Proof & burst pressures
6, 7, 8	Fungus resistance Leakage Proof & burst pressures
9	Ozone resistance

4.3 Verification methods. The types of verification methods included in this section are visual, inspection, measurement, sample tests, full-scale demonstration tests, simulation, modeling, engineering evaluation, component properties analysis, and similarity to previously approved or previously qualified designs.

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4.3.1 Verification alternatives. The manufacturer may propose alternative test methods, techniques, or equipment, including the application of statistical process control, tool control, or cost-effective sampling procedures, to verify performance. See the contract for alternatives that replace verifications required by this specification.

4.3.2 Inspection conditions. Unless otherwise specified (see 6.2), all inspections shall be conducted under the following conditions:

a. Air temperature: 77 ± 10°F
b. Barometric pressure: 28.5 (+2,-3) in. mercury.
c. Relative humidity: 50 ± 30%.

4.3.3 Operating requirements verifications. Complete each verification under 4.3.3.

4.3.3.1 Leakage. The hose shall be pressurized with air to the maximum working pressure as specified in table I for a period of 1 minute. A leakage rate greater than 0.02 ft^3/min, per in. of inside diameter (ID), per ft of length shall constitute failure of this test.

4.3.3.2 Proof and burst pressures. The hose shall be tested IAW ASTM D380 at the proof and burst pressures specified in table II. A rupture of the hose shall constitute failure of this test.

4.3.3.3 Vacuum resistance. A 36-in. length of hose shall be assembled with end couplings. One end shall be closed in such a manner as to prevent leakage of air, and the other end shall be connected to a vacuum pump. The hose shall then be subjected to a vacuum of 10 in. of mercury while curved to a radius equal to five times the OD of the hose for a period of 5 minutes. Subsequent to testing, a reduction of the hose OD greater than 5%, or a collapsed or separated hose ID, shall constitute failure of this test.

4.3.3.4 Flexibility. The hose shall be marked at four points spaced 90° apart around the external perimeter. The hose shall then be bent successively on each of the four marked points at 180° around a rigid cylinder with a diameter equal to the ID of the hose. Subsequent to testing, evidence of permanent kinking, flattening, fracturing, or deformation in excess of 2% shall constitute failure of this test.

4.3.3.5 Collapsing resistance. A 6-in. longitudinal section of a 36-in. test specimen shall be subjected to a load of 50 lb minimum, applied within 15 seconds. The load shall be applied perpendicular to the hose axis for a period of at least 5 minutes. When the load is removed, evidence of permanent setting, distortion, or deformation in excess of 5% shall constitute failure of this test.

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4.3.4 Interface requirements verification. Complete each verification under 4.3.4.

4.3.4.1 Envelope and interface dimensions. Use one or more of the methods outlined in 4.3.1 to verify that the envelope and interface dimensions are IAW applicable drawings (see 6.2).

4.3.4.2 Interchangeability. Use one or more of the methods outlined in 4.3.1 to verify that interchangeability between different assemblies having the same military part number is assured.

4.3.5 Support and ownership requirements verification. Unless otherwise specified (see 6.2), visually verify that the hose is permanently and legibly marked with the following information:

a. Nomenclature “HOSE, AIR DUCT”.
b. Military part number.
c. Manufacturer’s brand or firm name, or CAGE number.
d. Manufacturer’s serial number.
e. Date of manufacture (month and year).
f. Contract or order number.
g. Symbol “US”.

4.3.6 Operating environment requirements verification. Complete each verification under 4.3.6.

4.3.6.1 High temperature resistance. The hose shall be subjected to an internal flow of air at the applicable maximum temperature for its class (see 1.2.2). The air flow shall be maintained for a period of 72 hours with the ambient temperature at room temperature. Subsequent to testing, evidence of brittle material, cracks, or breaks shall constitute failure of this test.

4.3.6.2 Low temperature resistance. The hose shall be tested IAW ASTM D380 at a temperature of -65°F. Subsequent to testing, evidence of delamination, cracks, or breaks shall constitute failure of this test.

4.3.6.3 Vibration resistance. The hose shall be subjected to a constantly-applied vibration, with a double amplitude of 0.060 in. and a frequency cycled between 10 and 55 Hz in one-minute cycles. The vibration shall be parallel to the hose axis for 50 hours, and then perpendicular to the hose axis for another 50 hours. Subsequent to testing, evidence of delamination, cracks, or breaks shall constitute failure of this test.

4.3.6.4 Fungus resistance. The hose shall be tested IAW ASTM G21. Evidence of fungal growth shall constitute failure of this test. For first article testing, sample number 6 shall be tested for 30 days, number 7 for 60 days, and number 8 for 90 days. At the completion of the

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90-day test, sample number 8 shall be subjected to, and pass, the tests as specified in 4.3.3.1 and 4.3.3.2.

4.3.6.5 Ozone resistance. The hose shall be subjected to the applicable test as specified in ASTM D1171. Subsequent to testing, evidence of cracks or breaks shall constitute failure of this test.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of material is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Department of Defense Agency, or within the Military Department's System Command. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The hose covered by this specification is intended for use in heat-transfer applications in military vehicles. The hoses covered by this specification are military unique because they must be able to operate satisfactorily at ambient temperatures ranging from -65°F to +600°F. Commercial components are not designed to withstand such extreme environmental conditions and would experience failure.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Type and class of the hose required (see 1.2).
c. Issue of DoDISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.2).
d. If first article inspection is required (see 3.1 and 6.4).
e. Specification of interface drawing(s) (see 3.4.1 and 4.3.4).
f. If identification marking is other than as specified (see 3.5 and 4.3.5).
g. Specification of contractual sampling plan (see 4.1.2 and 6.5).
h. If inspection conditions are other than as specified (see 4.3.2).
i. Packaging requirements (see 5.1).
j. Part or identifying number (see 6.3).

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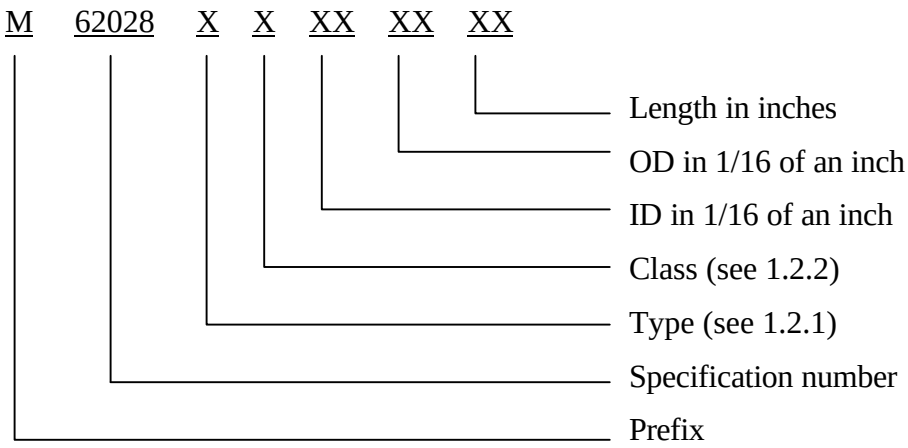
6.3 Part or Identifying Number (PIN). The PINs to be used for hoses acquired to this specification are created as follows:



Example: M6202823010572

Signifies: Type II, class 3, ID = 1/16 in., OD = 5/16 in., length = 72 in.

6.4 First article. When requiring a first article inspection, contracting documents should provide specific guidance to offerors. This guidance should cover whether the first article is a first article sample, a first production item, or the number of test items. These documents should also include specific instructions regarding arrangements for examinations, approval of first article test results, and disposition of first articles. Pre-solicitation documents should provide Government waiver rights for samples for first article inspection to bidders offering a previously acquired or tested product. Bidders offering such products who wish to rely on such production testing must furnish evidence with the bid that prior Government approval is appropriate for the pending contract.

6.5 Conformance inspection. Affordable conformance inspection with confidence varies depending upon a number of procurement risk factors. Some of these factors include: Contractor past performance, government schedules and budget, product material and design maturity, manufacturing capital equipment and processes applied, the controlled uniformity of those processes applied, labor skill and training, and the uniformity of measuring processes and techniques. During the solicitation, contracting documents should indicate those tests desired from table III and their designated frequency based on a risk assessment for the procurement.

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6.6 Subject term (key word) listing.

Crew compartments
Engine compartments
Heat transfer
Preheating

6.7 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Custodians:
Army - AT
DLA - CC

Preparing Activity:
Army - AT

(Project 4720-0317)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.

2. The submitter of this form must complete blocks 4, 5, 6, and 7, and send to preparing activity.

3. The preparing activity must provide a reply within 30 days from receipt of the form.

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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-PRF-62068D	2. DOCUMENT DATE *(YYYYMMDD)* 20011214

3. DOCUMENT TITLE HOSE, AIR DUCT: HIGH-TEMPERATURE FLEXIBLE, REINFORCED

4. NATURE OF CHANGE ***(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)***

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME *(Last, First, Middle Initial)*	b. ORGANIZATION	
c. ADDRESS *(Include Zip Code)*	d. TELEPHONE *(Include Area Code)* (1) Commercial (2) DSN *(If applicable)*	7. DATE SUBMITTED *(YYYYMMDD)*

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE *(Include Area Code)* (1) Commercial (586) 574-8745 (2) DSN 786-8745
c. ADDRESS *(Include Zip Code)* Commander U.S. Army Tank-automotive and Armaments Command ATTN: AMSTA-TR-E/AEIT 6501 E. 11 MILE ROAD Warren, MI 48397-5000	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Standardization Program Office (DLSC-LM) 8725 John J. Kingman Road, Suite 2533 Fort Belvoir, Virginia 22060-6221 Telephone (703) 767-6888 DSN 427-6888

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